CLINICAL · EAR

Wax, foreign bodies and canal stenosis

Wax and foreign bodies are among the commonest problems in the ear clinic, and among the easiest to make worse. Ask about a previous perforation before syringing, never chase a smooth round object deeper with forceps, kill a live insect before removing it, and treat a button battery as an emergency.

AT A GLANCE

- Ask about perforation before syringing.
- Never grasp a smooth object with forceps.
- Kill live insects first.
- Use a microscope and anaesthesia for impacted objects.
- Exostoses are multiple and bilateral (cold water); osteomas single at suture lines; never irrigate a button battery.

1 CHAPTER ONE Overview

The canal and what gets stuck in it.

1.1 Applied anatomy

The external auditory canal is about 24 mm long. Its outer third is cartilaginous, with hairs and sebaceous and ceruminous glands; its inner two-thirds is bony, with thin, sensitive skin. The canal narrows at the *isthmus*, about 6 mm from the drum: objects pushed past it become impacted. The *anterior recess*, in front of and below the drum, collects debris.

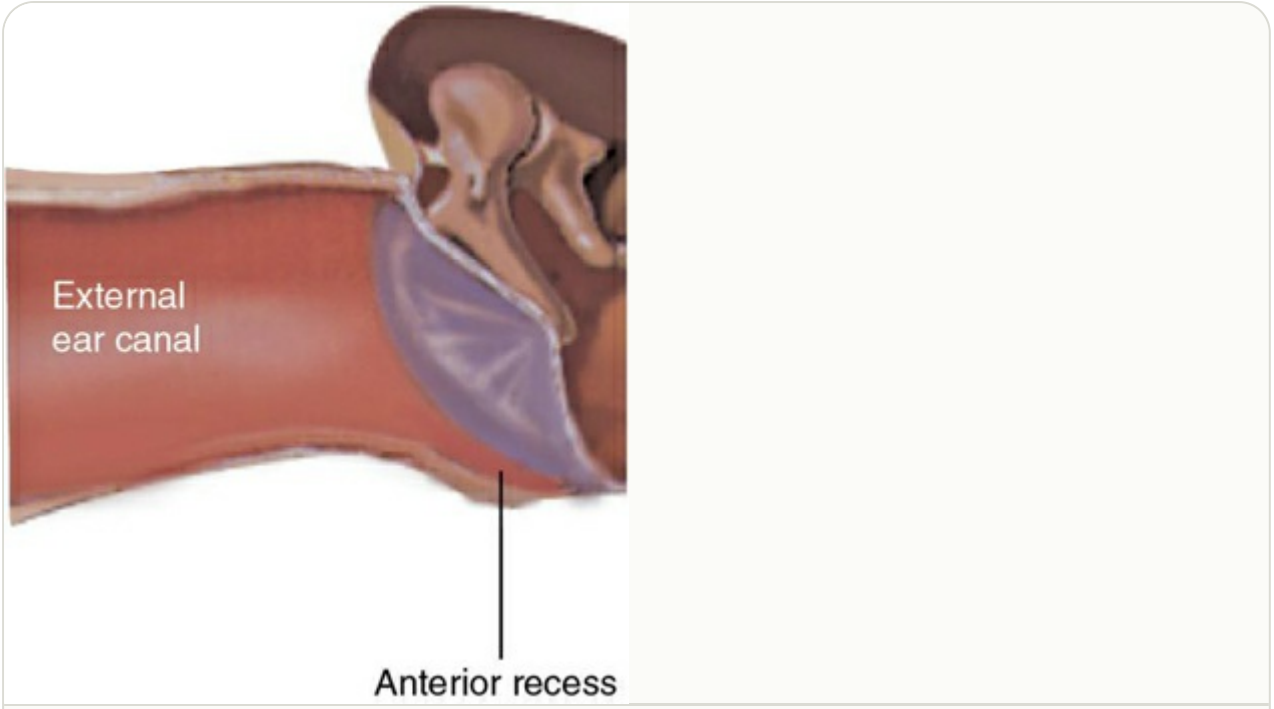


FIGURE 1 The anterior recess of the canal, where discharge and debris collect.
Dhingra, fig. 1.2

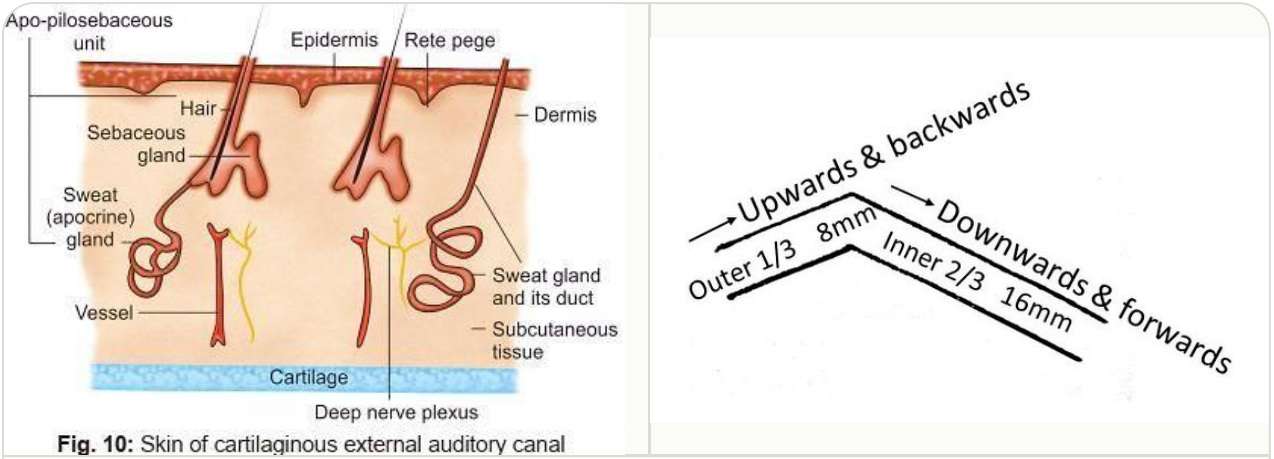


FIGURE 2 Skin of the cartilaginous canal, and the direction of the canal.
Anatomy atlas · ENT Steps p. 31, Doss p. 184

1.2 Pulling the pinna

In adults, pull the pinna up and back to straighten the canal; in infants, down and back.

2 CHAPTER TWO Wax

A useful secretion.

2.1 What it is

Wax (cerumen) is a mixture of secretions of the ceruminous and sebaceous glands with shed keratin. It is acidic, water-repellent and antibacterial, and the canal normally clears it by epithelial migration.

2.2 Why it impacts

A narrow or tortuous canal, stiff hairs, exostoses, overactive glands, and pushing wax deeper with cotton buds or hearing aid moulds.

2.3 Clinical picture

A blocked ear and hearing loss, sometimes tinnitus or giddiness, and occasionally a reflex cough through the auricular branch of the vagus (*Arnold's nerve*). Sudden blockage after bathing happens when water swells dry wax.

2.4 Removal

2.4.1 Softening

Hard wax is softened first with sodium bicarbonate glycerine drops, olive or almond oil, or hydrogen peroxide for a few days.

2.4.2 Syringing (irrigation)

Pull the pinna up and back and direct a stream of water at body temperature along the posterosuperior canal wall, so that it flows behind the wax and pushes it out. Dry the ear and inspect the drum afterwards.

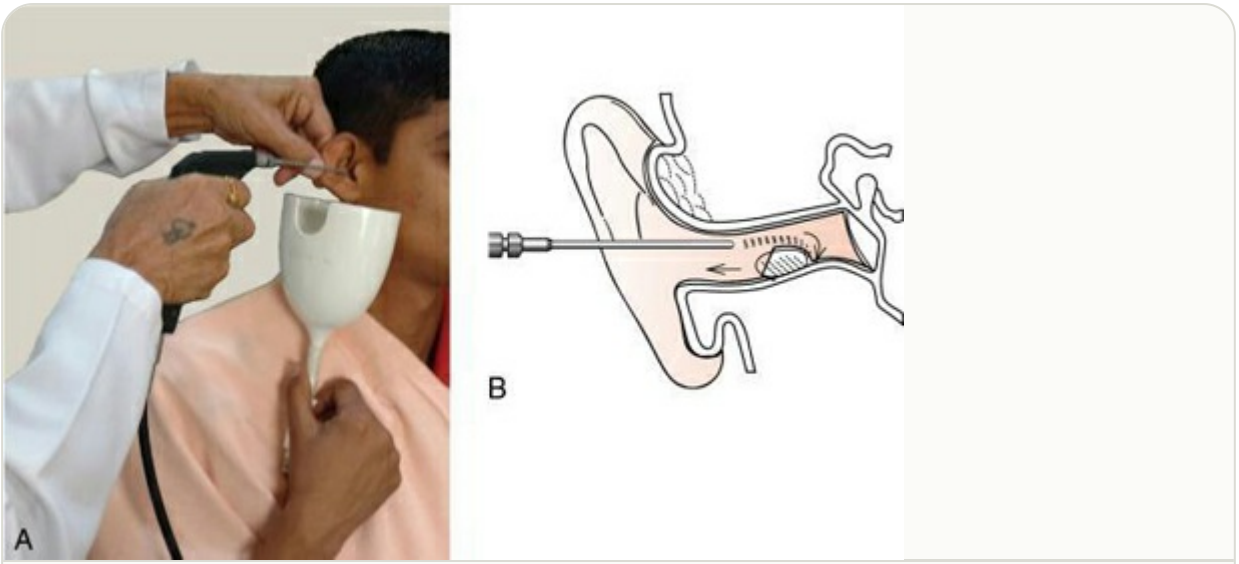


FIGURE 3 (A) Irrigation of the ear canal. (B) The jet passes behind the wax or foreign body and pushes it out.

Dhingra, fig. 8.13

DO NOT SYRINGE WHEN

- There is a perforation, or a history of discharge or ear surgery.
- The patient has had a previous perforation that may be thin.
- The foreign body is vegetable matter that will swell.
- The patient has an only hearing ear (relative).

2.4.3 Under the microscope

Suction or instruments (a wax hook, Jobson-Horne probe or crocodile forceps) under direct vision, when syringing is unsuitable.

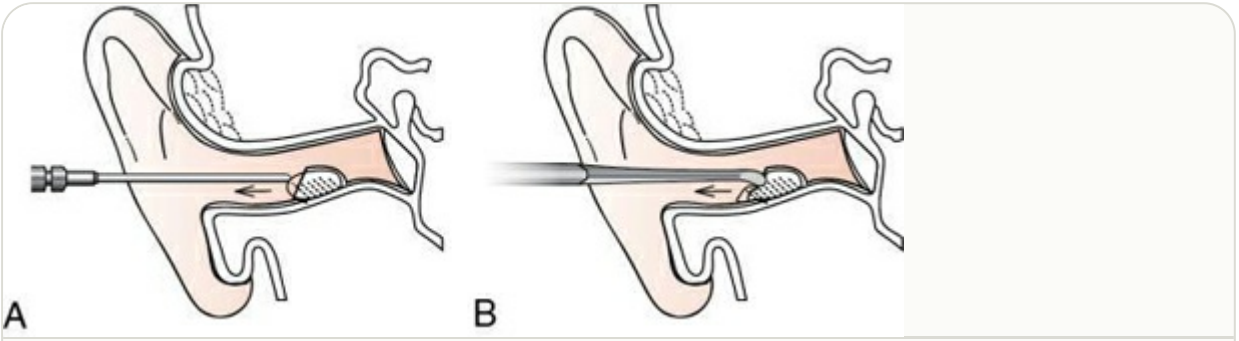


FIGURE 4 Other methods of removal: (A) suction and (B) forceps.

Dhingra, fig. 8.14

2.5 Complications of removal

Perforation, canal laceration and bleeding, otitis externa, and vertigo if the water is not at body temperature.

3 CHAPTER THREE Foreign bodies

What children and insects leave behind.

3.1 Types

3.1.1 Non-living

Beads, stones, paper, cotton, seeds, peas and beans, pencil tips, and button batteries. Vegetable matter swells with moisture and can cause infection.

3.1.2 Living

Insects (flies, cockroaches, ants), which cause intense discomfort and noise, and maggots, from flies laying eggs in a foul-discharging ear.

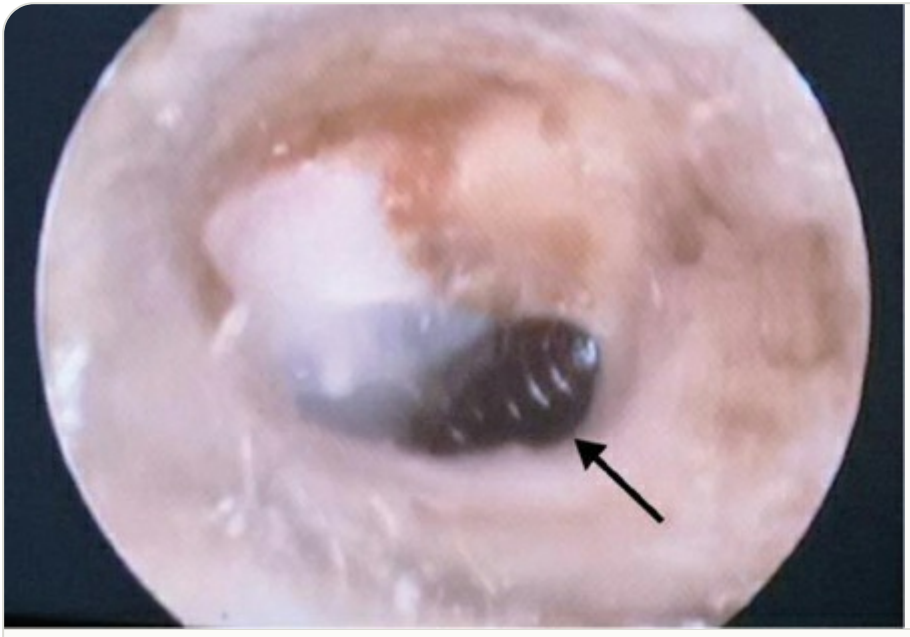


FIGURE 5 An insect in the ear canal, seen with an endoscope (arrow).
Dhingra, fig. 8.15

3.2 Removal

3.2.1 General principles

Use good light, the microscope and the right instrument. A cooperative patient, or a general anaesthetic for children and for impacted objects or a failed first attempt.

3.2.2 Smooth, round objects

Do not grasp with forceps: they slip and are pushed deeper past the isthmus. Use a hook passed beyond the object, suction, or syringing (if not vegetable).

3.2.3 Insects

Kill first by filling the canal with oil or spirit, then remove by forceps or syringing.

3.2.4 Maggots

Instil chloroform water or turpentine to kill them, then remove with forceps over several sittings; do not syringe. Treat the underlying ear disease.

3.2.5 Button batteries

Remove urgently: they leak alkali and cause rapid tissue necrosis. Do not instil drops or water.

3.2.6 Impacted deep objects

A post-aural approach may be needed for objects medial to the isthmus.

PEARLS

- Most foreign bodies can wait for the right setting; a bad first attempt makes the second harder.
- Always inspect the drum and the other ear (and the nostrils in a child) afterwards.

OPERATIVE GUIDES IN THEATRE 3 operations

Foreign Bodies of the External Auditory Canal
Foreign bodies of the canal

Office-Based Procedures in Otology
Office ear procedures

Congenital and Acquired Atresia and Stenosis of the External Auditory Canal
Atresia and stenosis

4 CHAPTER FOUR Canal stenosis

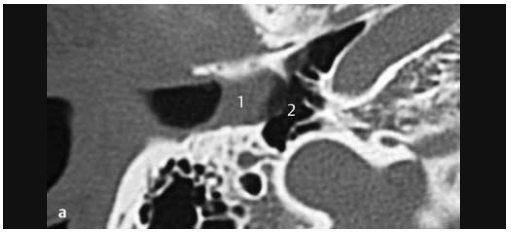
Narrowing after injury.

4.1 Causes

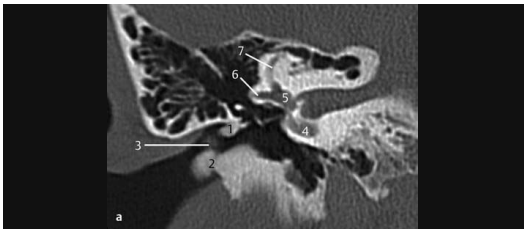
Acquired from chronic otitis externa, trauma, burns, surgery or radiation; congenital stenosis and atresia are covered separately.

IMAGING IN THE APP 2 cases

Canal stenosis and exostoses.



Stenosis of the External Auditory Canal
Bone disease · 2 images



Exostoses of the External Auditory Canal
Bone disease · 2 images

4.2 Treatment

Meatoplasty with excision of the fibrous tissue and skin flaps or grafts, and canal stenting while it heals.

EXOSTOSES, OSTEOMAS AND FOREIGN BODIES (OXFORD TEXTBOOK)

Exostoses are broad-based, usually multiple and bilateral bony swellings deep in the bony canal, caused by repeated exposure to cold water and wind ("surfer's ear"). **Osteomas** are single, pedunculated lesions arising at the tympanomastoid or tympanosquamous suture lines, unrelated to cold water. Both are usually symptomless but can trap wax and debris, causing recurrent otitis externa, a conductive loss and, rarely, canal cholesteatoma. Symptomatic patients are managed with regular microsuction and drops, and surgery is offered when that fails: a postauricular or endaural approach with a posterior meatal skin flap and removal of bone with a drill or chisel, protecting the canal skin, the drum and, posteriorly, the

facial nerve. Cold-water ear protection reduces recurrence. In a child, an apparent bony stenosis may be a mild congenital atresia, so CT is advised before operating.

Foreign bodies in the canal are removed under direct vision. Graspable objects are taken with forceps; round or smooth objects are hooked from behind with an angled hook, curette or suction rather than grasped, because forceps push them towards the drum. Irrigation is not recommended in the ENT clinic, and must never be used for vegetable matter, which swells, or for button batteries, which need urgent removal because they cause liquefactive necrosis.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 On call

SOMETHING STUCK IN THE EAR

1. Button battery: remove urgently, no drops.
2. Live insect: kill it first with oil or spirit, then remove.
3. Smooth round object: hook or suction, never forceps.
4. Impacted object or failed attempt: microscope, and consider general anaesthesia.
5. Inspect the drum afterwards.

5.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Anatomy:** cartilaginous and bony parts, the isthmus, the anterior recess.
- **Technique of syringing** and its contraindications.
- **Removal of foreign bodies,** living and non-living.

- **Arnold's nerve** and the ear cough reflex.

5.3 Common pitfalls

COMMON PITFALLS

- Syringing an ear with a perforation.
- Grasping a bead with forceps.
- Syringing vegetable matter.
- Delaying removal of a button battery.

Written for this app from Oxford Textbook of Otolaryngology ch. 13, Oxford Specialist Handbook ch. 15, Oxford Handbook of ENT ch. 5, Dhingra ch. 1, 8 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

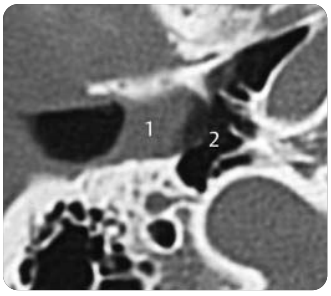
On imaging

The imaging of this topic: key findings, differentials and representative scans.

EXTERNAL EAR & TYMPANIC MEMBRANE · BONE DISEASE

Stenosis of the External Auditory Canal

- Beware of accumulation of cerumen or epithelium behind the fibrous wall, and exclude deeper pathology in the middle ear. In most cases, the fibrous wall can be surgically separated from the inner layers of the eardrum without opening the middle ear.
- In cases of malignant external otitis or secondary cholesteatoma, deeper pathologies (e.g., to the facial nerve) are of special interest.

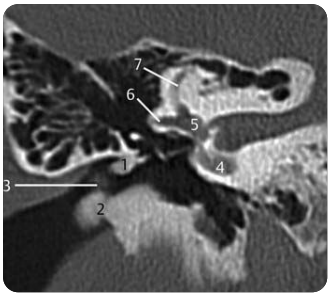


CT, axial. Chronic external otitis may lead to cicatrizing fibrous changes in the wall (1) of the external auditory canal. In this patient, the fibrous wall i…

EXTERNAL EAR & TYMPANIC MEMBRANE · BONE DISEASE

Exostoses of the External Auditory Canal

- In patients with aural discharge, chronic otitis may be the result of infected epithelial stasis.
- In cases with a narrowed orifice and meatus, a meatoplasty might be considered for better aeration and cleaning options. Furthermore, CT might be reassuring, showing a normal aerated middle ear.



CT, axial. Small exostoses are visualized in the roof of the external auditory canal near the annulus (1), and a larger one laterally on the floor …